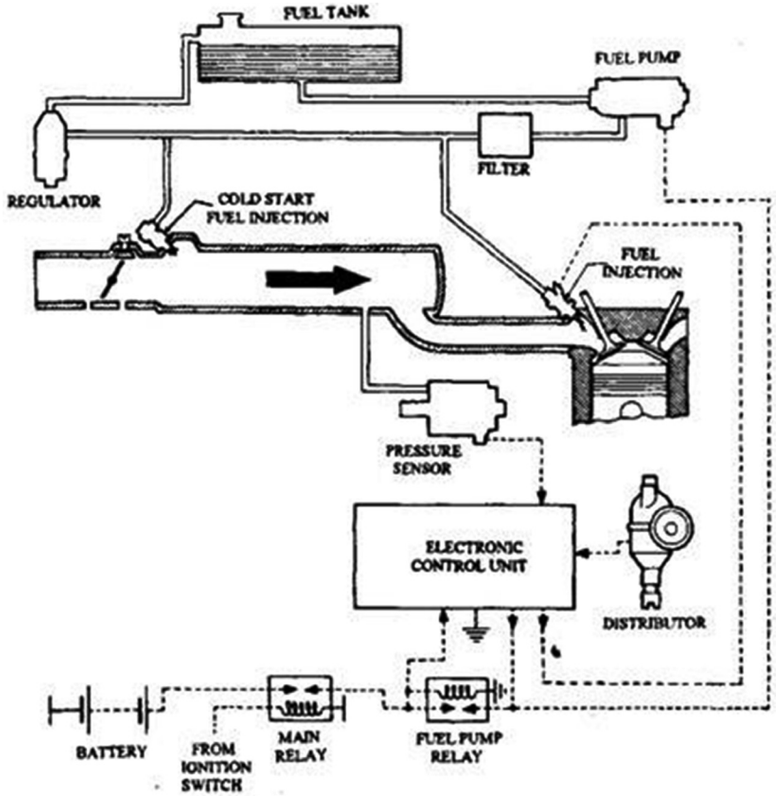


Figure 9.6 Layout of Electronic Fuel Injection System

In this fuel injection system electrically operated control valve with injection pump is introduced. This injection system receives petrol fuel from the fuel tank and injects at high pressure to the engine through metering distributor.

Inside the injector there is a solenoid which is operated under electrical supply with the help of electron control unit. ECU is a preprogrammed control unit with computer.

The injection of petrol fuel timing was given to the injector through the ECU with help of signals from the sensors. Meter valve receives the signal from the sensor and controls the solenoid for injecting petrol fuel in proper ratio in the each cylinder.

The signal quality received from the ECU shows need of some other purpose of the engines behaviour. To determine such behaviours so many sensors are introduced in to the engine. The behaviours are finally converted and send to the ECU. The required sources are air intake temperature, engine load, engine pressure, engine performance, air flow rate, engine coolant temperature, oxygen sensor, etc.,

9.6.2 Types of Petrol Injection

1) Direct injection
2) Single point or throttle body injection
3) Multi point or port injection
4) Timed and continuous fuel injection.

a) Combustion chamber injection (or) direct injection:

It will looks like diesel engine. In this type of injection, it was designed to inject petrol fuel directly to the engine cylinder.

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